

Project Title
Library Management
System

Introduction to
Problem Solving
(CSE1021)

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1.Introduction

The Movies & Shows Management System is a Python-based console application designed to organize and manage personal entertainment content. Users can add movies, add TV shows, search, delete, and mark items as watched. The system uses JSON storage for persistent data.

The project demonstrates Python fundamentals such as functions, conditionals, loops, and file handling.

2. Problem Statement

.People often struggle to keep track of what movies and shows they have watched, plan to watch, or want to remember later. There is no simple offline system for maintaining lists, checking watch status, or organizing content.

This project solves this problem by creating a simple, menu-driven management tool.

3. Functional Requirements

Add Movie

Add Show

View all items

Mark as watched

Delete item

Search by title keyword

Save and load data using JSON

User-friendly menu interface

4. Non-Functional Requirements

Usability: Simple and intuitive CLI menu

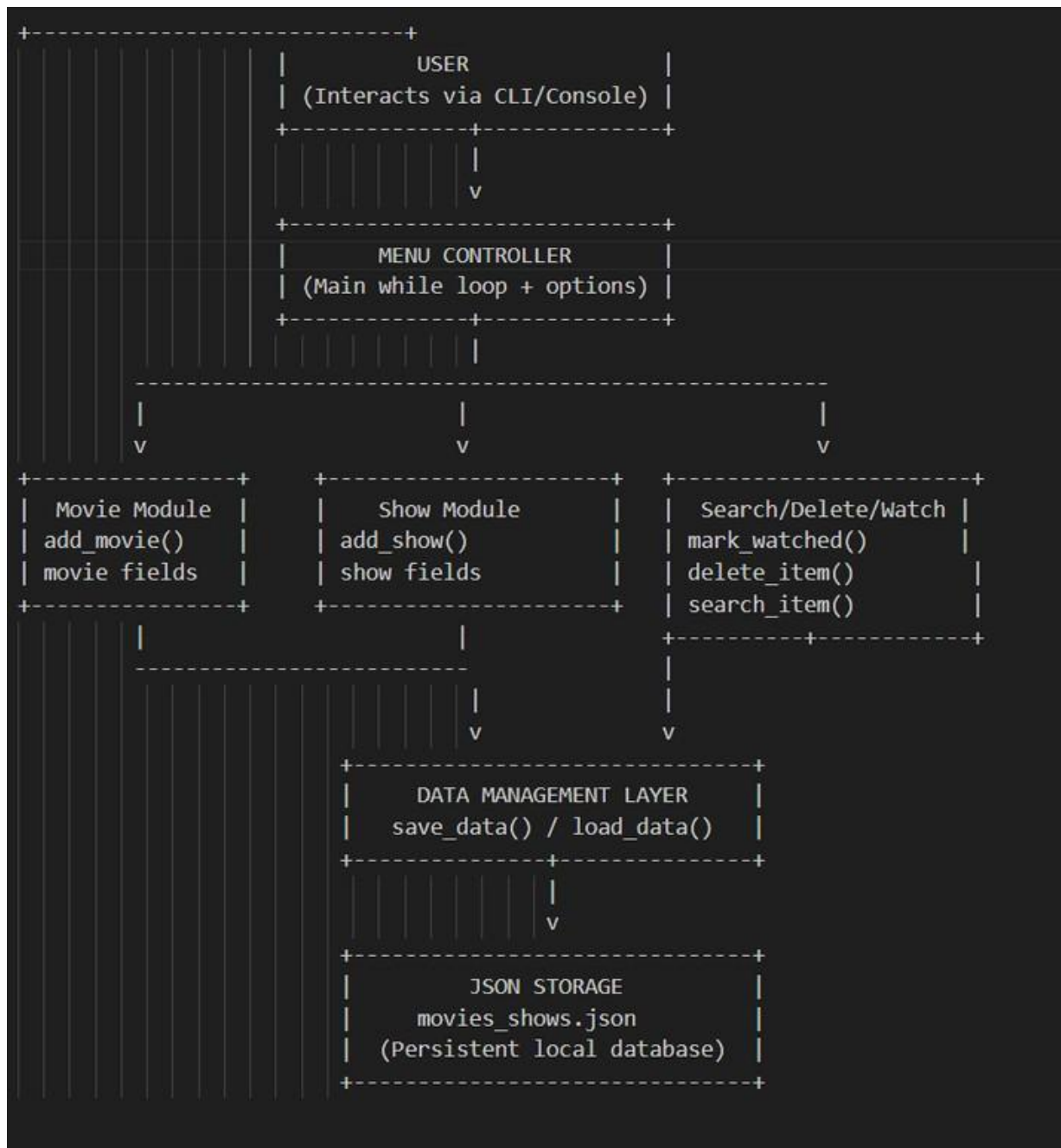
Reliability: Data stored in JSON format persistently

Portability: Works on Windows, Linux, Mac, and Android (Pydroid/Termux)

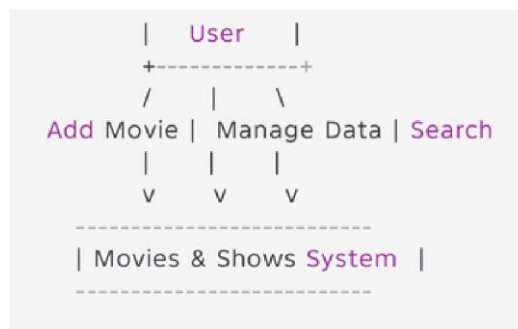
Performance: Fast response, low memory usage

Scalability: Easy to add future

5. System Architecture



7. Design Diagrams



a) Use Case Diagram

b) Workflow Diagram



c) Sequence Diagram

```
User → System: Select Option
System → User: Ask for Inputs
User → System: Provide Inputs
System → File: Save/Load JSON
File → System: Return Data
System → User: Show Output
```

d) Class/Component Diagram

```
+-----+
| MoviesShowsSystem |
+-----+
| - data (list)      |
+-----+
| + add_movie()      |
| + add_show()       |
| + delete_item()    |
| + mark_watched()   |
| + search_item()    |
| + list_items()     |
+-----+
```

e) ER Diagram (if storage used)

Movie

title
year
duration
genre
watched

Show

title
year
seasons
episodes
duration
watched

8. Design Decisions & Rationale

JSON file chosen for simple persistent storage

Functions used to modularize code

Menu-based design for ease of use

Separate Add Movie & Add Show for flexibility

Dictionary-based data structure for structured attributes

9. Implementation Details

Written in Python 3

Uses json module for file handling

Lists & dictionaries store movie/show information

While loop used for menu operations

Status updates (watched/pending) stored as boolean values

10. Screenshots / Results

```
=== Movies & Shows Management ===
1. Add Movie
2. Add Show
3. View All
4. Mark as Watched
5. Delete Movie/Show
6. Search
7. Exit
Choose an option: 1

--- Add Movie ---
Movie title: Baahubali
Release year: 2015
Duration (e.g. 2h 15m or 130 min): 150
Genre (Action, Drama, Sci-Fi, etc): Action
Movie added successfully!

=== Movies & Shows Management ===
1. Add Movie
2. Add Show
3. View All
4. Mark as Watched
5. Delete Movie/Show
6. Search
7. Exit
Choose an option: 2

--- Add Show ---
Show title: Stranger things
Release year: 2016
Number of seasons: 5
Total episodes: 8
Duration per episode (e.g. 45 min): 50
Show added successfully!
```

=== Movies & Shows Management ===

1. Add Movie
2. Add Show
3. View All
4. Mark as Watched
5. Delete Movie/Show
6. Search
7. Exit

Choose an option: 3

----- ALL MOVIES & SHOWS -----

1. [MOVIE] baahubali (2015) - 150 - action - Pending !
2. [MOVIE] Baahubali (2015) - 150 - action - Pending !
3. [MOVIE] Baahubali (2015) - 150 - Action - Pending !
4. [SHOW] Stranger things (2016) - 5 Seasons, 8 Episodes - 50 - Pending !

=== Movies & Shows Management ===

1. Add Movie
2. Add Show
3. View All
4. Mark as Watched
5. Delete Movie/Show
6. Search
7. Exit

Choose an option: 4

----- ALL MOVIES & SHOWS -----

1. [MOVIE] baahubali (2015) - 150 - action - Pending !
2. [MOVIE] Baahubali (2015) - 150 - action - Pending !
3. [MOVIE] Baahubali (2015) - 150 - Action - Pending !
4. [SHOW] Stranger things (2016) - 5 Seasons, 8 Episodes - 50 - Pending !

Enter item number to mark as watched: 1

Item marked as watched!

11. Testing Approach

Input Testing: Checked with valid/invalid inputs

Search Testing: Verified partial keywords

Delete Testing: Verified deletion of movies and shows

JSON File Testing: Ensured reading and writing works

Boundary Testing: Empty list, 1 item, multiple item

12. Challenges Faced

Handling JSON data formatting

Designing a clear and simple menu

Managing separate structures for movies and shows

Keeping the code modular and readable

13. Learnings & Key Takeaways

Understanding of JSON-based storage

Importance of modular programming

Real-world data structuring using dictionaries

Python file handling and loops

Project documentation and design diagrams

14. Future Enhancements

Add OTT platform (Netflix, Prime, Hotstar)

Add IMDB rating system

Add sorting & filtering (Year, Genre, A–Z)

Export data to Excel/PDF

Tkinter GUI version

Movie recommendations based on genre

15. References

Python Official Documentation (python.org)

W3Schools Python Reference

GeeksForGeeks – JSON Handling

Class lecture notes